

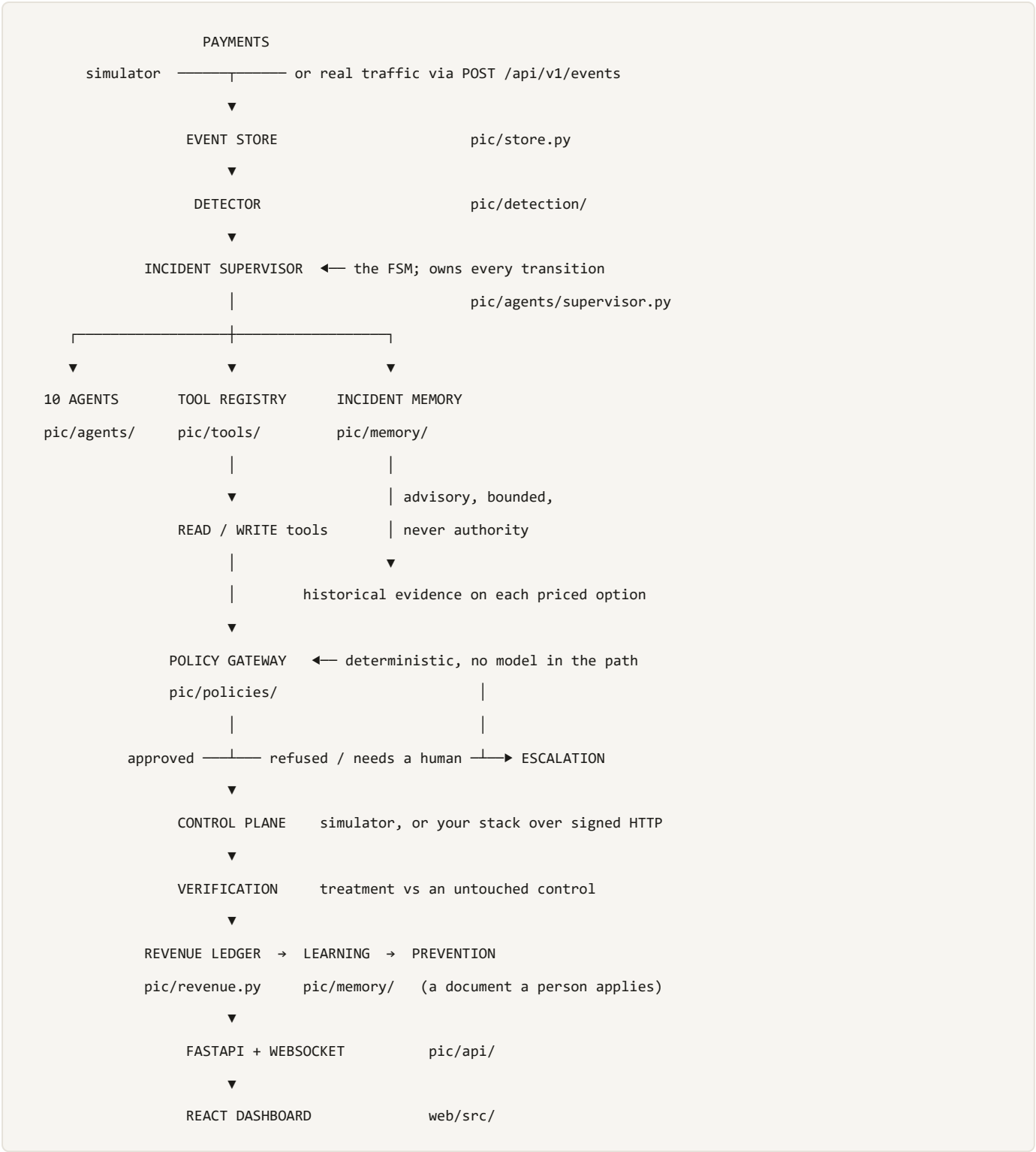
The project map — what is happening, and where

This document answers one question: if something is happening in this system, which file is doing it?

The other documents each take one cut. ARCHITECTURE is the contract, DECISIONS is why each part is shaped as it is, WALKTHROUGH follows a single payment failure through the code, EVALUATION is how well it works, INTEGRATION is how to point it at a real merchant, and DEMO is how to show it.

This one is the index. Every subsystem, every file, every screen, and what produces what.

1. The shape of it



Two rules that the whole design hangs off, and where each is enforced:

RULE	ENFORCED IN
The model can propose but never execute	pic/policies/gateway.py , pic/tools/registry.py::call

RULE	ENFORCED IN
History is evidence, never authority	<code>pic/agents/strategy.py</code> (bounded prior), asserted in <code>tests/test_learning_and_recovery.py</code>

2. Every directory, and what lives in it

pic/ — the system

PATH	LINES	WHAT IT DOES
<code>pic/schemas.py</code>	907	Every typed contract. If it crosses an agent boundary or gets persisted, it is defined here and nowhere else.
<code>pic/config.py</code>	186	Thresholds and tunables — detection windows, verification significance, speed-up factors.
<code>pic/engine.py</code>	411	Wires simulator + store + detector + tools + gateway + memory + supervisor. The demo, the API and the benchmark all construct through this, so none of them can diverge.
<code>pic/store.py</code>	497	In-memory event index. Timestamp-sorted with a parallel epoch array, so a window slice is a bisect not a scan.
<code>pic/database.py</code>	272	SQLAlchemy audit schema — incidents, agent steps, tool calls, policy decisions, audit records, incident memory, prevention. SQLite by default.
<code>pic/revenue.py</code>	169	The money. Turns per-hour rates into what an incident cost and saved, with the identity <code>at risk = protected + recovered + lost</code> true by construction.
<code>pic/demo.py</code>	634	The terminal demo, both acts.

pic/agents/ — one responsibility each

FILE	LINES	OWNS	PRODUCES
<code>pic/agents/supervisor.py</code>	1126	The FSM. Every state transition, the intervention cap, the clock, learning at close.	—
<code>pic/agents/base.py</code>	172	The agent contract: timing, tool-call capture, failure containment, step recording.	<code>AgentStep</code>
<code>pic/agents/detection.py</code>	30	Wraps the detector as an agent.	<code>AnomalySignal</code>
<code>pic/agents/investigation.py</code>	783	Evidence gathering, and separating real second faults from echoes of the first.	<code>EvidenceBundle</code>
<code>pic/agents/impact.py</code>	136	Prices the degradation per hour, showing every step of the arithmetic.	<code>ImpactAssessment</code>
<code>pic/agents/root_cause.py</code>	570	Scores hypotheses from evidence features; the model may re-rank but never invent.	<code>RootCauseAssessment</code>

FILE	LINES	OWNS	PRODUCES
<code>pic/agents/strategy.py</code>	968	Prices every way out , at several magnitudes, and attaches what happened last time each was tried.	<code>RecoveryPlan</code>
<code>pic/agents/decision.py</code>	180	Chooses one priced option on expected value. Cannot generate options.	<code>ActionProposal</code>
<code>pic/agents/action.py</code>	250	The only agent that calls a write tool, and only with a decision naming that action. Writes the audit record with its inverse.	<code>ActionResult</code> , <code>AuditRecord</code>
<code>pic/agents/verification.py</code>	438	Treatment against a concurrent control, significance-tested.	<code>VerificationResult</code>
<code>pic/agents/recovery.py</code>	214	The second layer: payments that already failed, under their own policy decision.	<code>OrderRecoveryResult</code>
<code>pic/agents/escalation.py</code>	466	Hands over with the case, not the category, and a list of moves a human can actually make.	<code>Escalation</code>

`pic/detection/` — why detection is not an LLM

FILE	LINES	WHAT
<code>pic/detection/detector.py</code>	512	Two-tier detector: a fast headline tier and a slower segment tier that catches what the headline hides. Three tests must agree.
<code>pic/detection/statistics.py</code>	163	EWMA baseline, robust z-score, CUSUM change point, two-proportion z-test, Wilson bounds.

`pic/memory/` — the closed loop

FILE	LINES	WHAT
<code>pic/memory/store.py</code>	664	Storage and deterministic retrieval. <code>find_similar_incidents</code> , <code>get_success_rate_for_action</code> , <code>get_action_outcomes_for_condition</code> , <code>get_historical_recovery_outcomes</code> . Merchant-scoped by default.
<code>pic/memory/build.py</code>	216	The only writer of memory . Copies fields from typed agent outputs; no model on the path.
<code>pic/memory/playbook.py</code>	360	The between-incidents view: preferred intervention per signature, what to avoid, effectiveness by method and provider. Read-only, off the decision path .
<code>pic/memory/patterns.py</code>	240	Mines recurring signatures into <code>PreventionRecommendation</code> s. Recommends; cannot apply.
<code>pic/memory/profile.py</code>	216	Merchant operating profile, derived from their policy and their own outcomes. No hand-written preferences.

pic/policies/ — the gate

FILE	LINES	WHAT
pic/policies/gateway.py	551	Deterministic rule evaluation. Most restrictive outcome wins, clamp before refusing, unknown means no.
pic/policies/merchant_policies.yaml	—	The merchant's standing grant of authority. Data, not code.

pic/tools/ — every fact comes from here

FILE	LINES	WHAT
pic/tools/registry.py	164	Typed registration, call recording, and the gate: a write tool without an approving decision raises .
pic/tools/read_tools.py	620	18 read tools. Pure functions of the store and a window.
pic/tools/write_tools.py	410	11 write tools. The only side-effecting code, each with a declared inverse.

pic/recovery/ , pic/simulation/ , pic/llm/ , pic/api/ , pic/integration/ , pic/evaluation/

FILE	LINES	WHAT
pic/recovery/orders.py	186	Which failed payments are still completable, by what means, at what measured probability.
pic/simulation/generator.py	654	Traffic generator and the <code>ControlPlane</code> the write tools mutate. Ground truth accumulates from the generative process.
pic/simulation/scenarios.py	328	Nine injectable degradations with known ground truth.
pic/simulation/seed_memory.py	195	Deterministic seeded history. Every record flagged <code>seeded=True</code> .
pic/llm/base.py	97	The reasoner interface — three narrow methods, deliberately.
pic/llm/gemini.py	352	Gemini client, with validation against the evidence bundle and catalogue.
pic/llm/deterministic.py	100	The fallback that makes the repo run with no credentials.
pic/api/main.py	883	FastAPI backend, per-visitor simulation sessions, WebSocket stream.
pic/api/trace.py	462	Builds the explainable decision trace, stage by stage.
pic/api/v1.py	237	The authenticated integration API for real merchants.
pic/integration/ingest.py	103	Validating real payment events; unknown status is rejected, not guessed.
pic/integration/control.py	242	The webhook control plane. The merchant's response is the source of truth.
pic/integration/signing.py	61	HMAC request signing.
pic/integration/tenant.py	218	Per-merchant engines, keys and policy files.
pic/evaluation/harness.py	1065	The benchmark. Every published number comes from here.

web/src/ — the dashboard

FILE	LINES	WHAT
web/src/App.jsx	384	Routing, polling cadences, the action handlers.
web/src/Incident.jsx	736	The incident story. Hero → recommendation → options → verification → money → learning → prevention → trace.
web/src/pages.jsx	699	Overview, Incidents list, Learning, Health, How it works.
web/src/components.jsx	638	Shared primitives, the trace renderer, the money bar, record rows.
web/src/Simulate.jsx	176	Scenario injection and custom incidents.
web/src/Landing.jsx	249	The front door.
web/src/Chart.jsx	207	The success-rate chart.
web/src/api.js	183	The client, session handling, and all money/duration formatting.
web/src/styles.css	1392	One accent, warm paper, hierarchy by size not colour.

tests/ — 146 tests

FILE	LINES	GUARDS
tests/test_safety_invariants.py	671	Ground truth unreachable, write tools gated, policy outcomes, override refusals, handover quality.
tests/test_learning_and_recovery.py	1053	Memory, learning, revenue identity, order recovery, prevention, and safety under learning .
tests/test_detection_and_simulation.py	370	Statistics, control-plane invertibility, segment maths, detection quality.
tests/test_end_to_end.py	320	Full lifecycles: happy path, failed intervention, restraint, echoes.
tests/test_integration_api.py	315	The real-merchant path over HTTP, signing, replay, read-only tenants.

3. The lifecycle — 17 states, and who runs in each

pic/schemas.py::IncidentState , driven by pic/agents/supervisor.py::run_incident .

OBSERVING → DETECTED → INVESTIGATING → IMPACT_ASSESSED → DIAGNOSING

→ RECOVERY_PLANNING → DECIDING → POLICY_REVIEW

→ EXECUTING → VERIFYING → RESOLVED → RECOVERING_ORDERS → LEARNING → CLOSED

→ AWAITING_HUMAN_APPROVAL ─┐

→ ROLLING_BACK → ESCALATED → LEARNING → CLOSED

STATE	AGENT	PRODUCED	FAILURE BEHAVIOUR
DETECTED	detection	AnomalySignal	No signal → no incident (fail closed)
INVESTIGATING	investigation	EvidenceBundle	Partial bundle, confidence capped
IMPACT_ASSESSED	impact	ImpactAssessment	Escalate — never guess money
DIAGNOSING	root_cause	RootCauseAssessment	Wait and re-investigate, then escalate
RECOVERY_PLANNING	recovery_strategy	RecoveryPlan	Decision agent prices the same set; history is lost, the response is not
DECIDING	decision	ActionProposal	Escalate
POLICY_REVIEW	(the gateway, outside any agent)	PolicyDecision	Deny → escalate; require approval → park
EXECUTING	action	ActionResult + audit	Rollback, escalate
VERIFYING	verification	VerificationResult	Treat as failed (conservative)
ROLLING_BACK	action.rollback	audit record	Escalate rollback_failed
RECOVERING_ORDERS	order_recovery	OrderRecoveryResult	Records that nothing ran, and why
LEARNING	(supervisor)	RevenueOutcome + IncidentOutcomeRecord	—
ESCALATED	escalation	Escalation	Terminal, with next steps

Invariants, all asserted in tests: EXECUTING is reachable only from an approved PolicyDecision; every transition appends an immutable AgentStep; the attempt cap forces escalation rather than thrashing; RECOVERING_ORDERS is skipped when the fix was reverted; and LEARNING runs for every closed incident, including the ones that went badly.

4. The API surface, and what produces each response

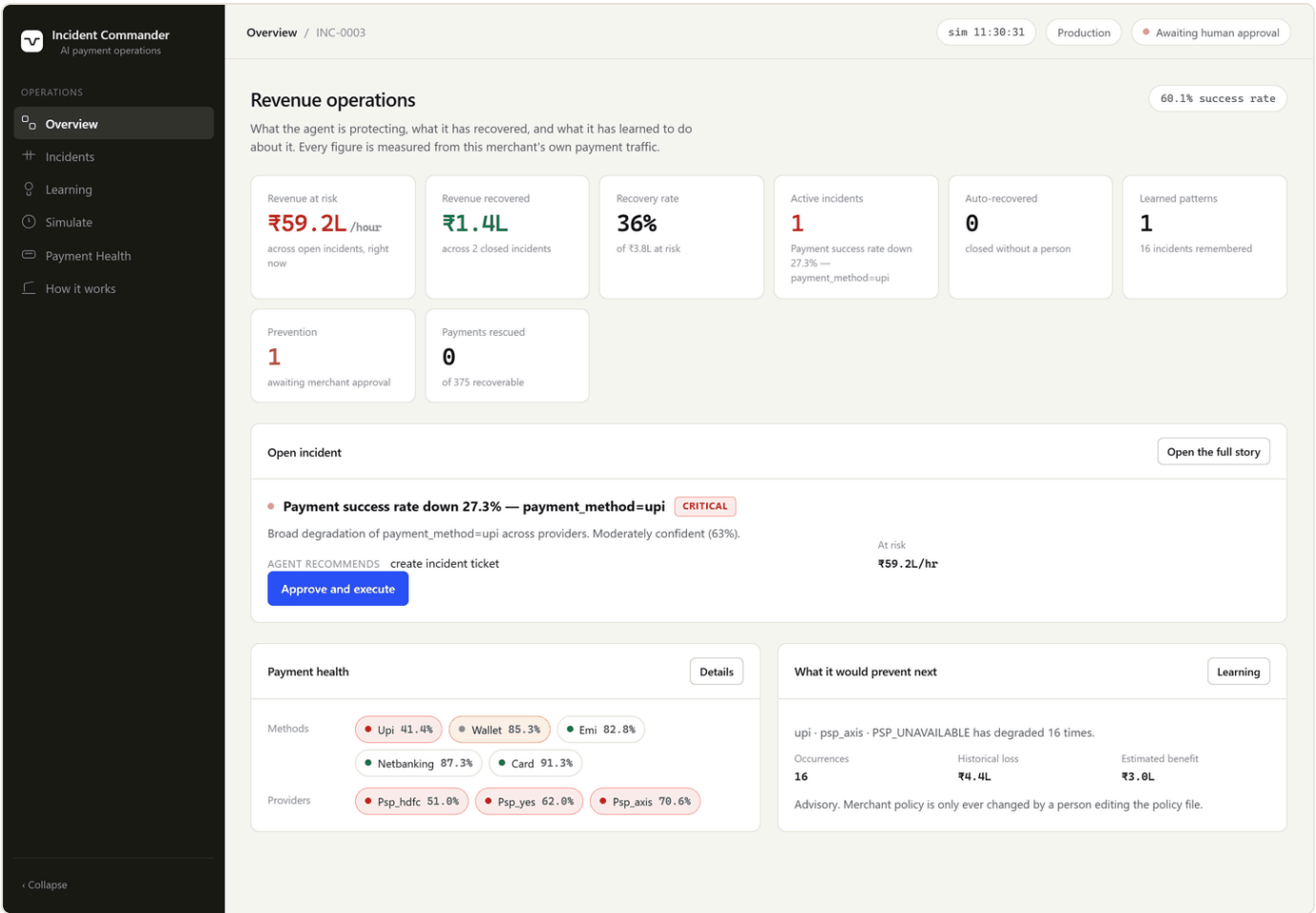
ENDPOINT	PRODUCED BY
GET /api/metrics	Engine.current_metrics — window stats, revenue.py aggregate, memory counts
GET /api/incidents	The supervisor's incident list
GET /api/incidents/{id}	IncidentRecord.model_dump — the raw typed record
GET /api/incidents/{id}/trace	api/trace.py::build_trace — 14 stages with their evidence
POST /api/incidents/{id}/approve reject override retry acknowledge	supervisor.approve / reject / override / retry / acknowledge

ENDPOINT	PRODUCED BY
GET /api/learning	Engine.learning_summary — memory, playbook.py , effectiveness, prevention
GET /api/prevention	supervisor.prevention , mined by patterns.py
POST /api/prevention/{id}/{accept\ dismiss}	supervisor.acknowledge_prevention — records; changes no policy
POST /api/demo/seed-history	simulation/seed_memory.py::seed
GET /api/policy	The merchant YAML in force
GET /api/health/segments	store.segment_stats against each segment's own baseline
GET /api/evaluation	evaluation/results/latest.json , written by the harness
POST /api/v1/events	integration/ingest.py — authenticated, per merchant
WS /ws	The supervisor's lifecycle events, fanned out

5. The UI, screen by screen

Everything below is a real screenshot of the running system, captured by driving it in a headless browser. Every figure in them came from a typed field the agents recorded.

5.1 Overview — the executive view



Eight cards answering, in five seconds: how much is at risk, how much came back, how much of that was automatic, what has been learned, what could be prevented.

REGION	COMES FROM
The eight cards	<code>GET /api/metrics</code> → <code>Engine.current_metrics</code>
<i>Revenue recovered, Recovery rate</i>	<code>revenue.py::aggregate</code> over closed incidents
<i>Auto-recovered</i>	Incidents whose learning record shows <code>helped</code> and no human involvement
<i>Learned patterns</i>	Distinct failure signatures in memory
<i>Prevention</i>	<code>patterns.py</code> , <code>status</code> <code>PROPOSED</code>
The incident glance	The selected incident's typed record; the bar is <code>RevenueBar</code>
Payment health chips	<code>GET /api/health/segments</code>

Rendered by `pages.jsx::CommandCenter` .

5.2 The incident story

Incident Commander

18 payment operations

OPERATIONS

Overview

Incidents

Learning

Simulate

Payment health

How it works

Incidents

18 payment operations

Incidents / INC-0003

11:52:11

Production

Awaiting human approval

Incidents

3 recorded · newest first

INC-0003

CRITICAL

Payment success rate down 27.2% — payment_method=upi

11h 20m

● need approval

INC-0002

CRITICAL

Payment success rate down 0.7% — payment_method=upi · psp=psp_axis

1h 31m

● escalated

INC-0001

CRITICAL

Payment success rate down 14.8% — payment_method=upi · psp=psp_axis

15h 7m

● partially recovered

INCIDENT CLOSED

INC-0003

CRITICAL

Payment success rate down 14.8% — payment_method=upi · psp=psp_axis

Payment service provider degradation on psp=psp_axis, 57% confidence.

THIS INCIDENT'S REVENUE AT RISK

₹3.4L

Lost the ₹3.4L in 7h 45s while it was open

Revenued

₹0

0% payments completed

Protected

₹1.4L

50% losses prevented

Recovery

40%

all users out of risk

Duration

7h 45s

open to closed

Payments received

0

of 275 accountable

AGENT RECOMMENDATION

SUGGESTED

Shift 15% of upi traffic from route_A to route_C

Expected revenue protected

₹9.7L/hour

Confidence in the cause

57%

Risk

Low

score 0.95

Historical success

8 / 9

comparable incidents improved

WHY THIS ACTION

- Payment service provider degradation on psp=psp_axis, at 57% confidence.
- Route route_A is at 34.2% success for upi while route_C is at 94.3%. Moving 20% of that traffic accounts roughly 1.6M payments/hour from the worse rate to the better one. 7 of 9 comparable incidents fully recovered with this action at 20%. 1 partially. Median recovery 4.2 min. Median revenue protected ₹10.2M. 1 required rollback.
- The action was proved at 20%. Merchant policy granted 15%, which is what runs — the expected value above is for the action as proved.
- ₹26.7L an hour is at risk, from about 2.5M extra failed payments affecting 2,662 customers an hour.
- 8 of 9 comparable incidents improved with this action at this risk.
- Merchant policy permits it only 15% of cumulative shift headroom remains on route_A, not-cause confidence 0.57 is below the autonomous floor 0.70 approved by operator.
- The 12 most similar of 14 comparable incidents.

WHAT HAPPENS IF THIS IS WRONG

- Only 15% of upi traffic moves. The next steps on route_A and is the control group.
- The result is measured against that control group before the incident can be called recovered, so the incident worsening on its own cannot be mistaken for the fix working.
- Nothing is judged for 180s: the system waits and watches real traffic rather than deciding success immediately.
- If it does not help, the recorded revenue is relogged automatically and the incident goes to a human.

APPROVE

only 15% of cumulative shift headroom remains on route_A, not-cause confidence 0.57 is below the autonomous floor 0.70 approved by operator

Merchant policy reduced to: requested from route route_A, to route route_C, percentage 20, granted from route route_A, to route route_C, percentage 15.

Every option is considered

0 priced · selected highlighted

STRATEGY

EXPECTED VALUE

PROTECT/HOUR

RISK

HISTORICAL SUCCESS

SELECTED

Shift Traffic 20%

route_A → route_C

₹9.4L

₹9.4L

Low

8/9 · 80%

Shift Traffic 15%

route_A → route_C

₹9.1L

₹7.1L

Low

8/9 · 80%

Create Incident Ticket

incident_tracker

₹0

₹0

Minimal

no history

Set Monitoring Frequency

observation_evidence

₹0

₹0

Minimal

no history

No Action

₹0

₹0

Minimal

no history

Expected value is revenue protected weighted by the probability the fix works, less what it risks if it backfires, history adjusts that probability within a hard bound: it cannot create an option, change a measured figure, or reach the policy gateway.

DID IT ACTUALLY WORK?

PARTIALLY RECOVERED

Control: left alone

26.3%

80 experiments

+69.7

percentage points

p < 0.0001

Treatment: moved

96.0%

228 experiments

The control group is traffic the agent deliberately did not move, being through the same ring and the same hour. Comparing against it is what stops the incident recovering on its own from being mistaken for the fix working.

Real improvement, but not the whole gap — the incident stayed open.

Traffic moved to route_C is converting at 96.2% against 26.3% for the 80 payments left on route_A (4487% p=0.0001). Measured against a concurrent control, so the incident's own trajectory cannot account for it. Control success rate is still 0.5% below baseline, so the incident stays open.

Revenue protected

₹26.7L/hour

What happened financially

every figure is scientific or ever recorded measurements

Protected

₹1.4L

Revenued

₹0

Lost

₹2.0L

Recovery rate

0%

Risk

₹1.4L

Protected

₹0

Initial payments

Recovery of 275 orders was not authorized: not-cause confidence 0.57 is below the autonomous floor 0.70: recovery method payment_link are not authorized for autonomous use and were dropped from the campaign. A person can authorize it from the incident.

The automatic, low by low

What the agent learned

written to memory when the incident closed

Recorded

shift traffic at 15% under this future pattern — partially recovered.

Agent signature

165 · psp_axis · PSP_UNAVAILABLE

Now stands at

8 of 9 comparable incidents improved with this action, 1 rolled back.

This will influence future recovery recommendations: it moves the efficacy prior behind every option of this kind, inside a hard bound. It cannot authorize anything — the policy gateway decides what may run, and history is evidence rather than authority.

See the learned playbook

Prevention opportunity

whenever · merchant policy is unchanged

PROPOSED

14 measurements · PSP-00000000

psi · psp_axis · PSP_UNAVAILABLE has degraded 16 times.

- when success rate down at least 1%
- when failure dominated by PSP_UNAVAILABLE
- when a healthy destination route was available in 14 of 14 cases

Desired

₹1.4L

Revenued loss

₹0

Estimated benefit

₹2.0L

Shift traffic from route route_A, to route route_C, percentage 15

The evidence behind this

Arbitrary. Applying this means a person editing the merchant policy file — approving here records the request and changes nothing.

Review it on the Learning page

Handed to a human

CRITICAL

Shift traffic is ready to run, but only 15% of cumulative shift headroom remains on route_A, not-cause confidence 0.57 is below the autonomous floor 0.70: it executes only if you approve it.

Approve or reject the proposed action in the dashboard.

WHAT NOW

Approve shift traffic

Policy asked for a person: only 15% of cumulative shift headroom remains on route_A, not-cause confidence 0.57 is below the autonomous floor 0.70. Run it, then measures the result against a control group and waits if it the numbers do not improve.

Run it

Reject it

Nothing runs. The incident stays on the record as declined by a human. Payments can still usually go through.

Do it

I have this

Results that a person took it with a note, and takes it off the board. Use it when the fix belongs somewhere else: a provider, a display, another team. Changes nothing about payments. The incident stays in the audit trail.

Do it

Agent trace

14 of 14 stages finished

Observation

Evidence

Hypothesis

Historical evidence

Candidate strategies

Decision

What happens if this is wrong

Policy check

Action

Control group

Result

Revenue outcome

Failed payment recovery

Learning

One argument, top to bottom. `web/src/Incident.jsx`.

The hero — what is breaking and what it costs

● INCIDENT CLOSED INC-0001

CRITICAL

Payment success rate down 14.8% — payment_method=upi · psp=psp_axis

Payment service provider degradation on psp=psp_axis. 57% confidence.

THIS INCIDENT'S REVENUE AT RISK

₹3.4L

over 7m 45s · ₹26.7L/hour while it was open

Protected ₹1.4L

Recovered ₹0

Lost ₹2.1L

Recovered	Protected	Recovery	Duration	Payments rescued
₹0	₹1.4L	40%	7m 45s	0
failed payments completed	further loss prevented	of what was at risk	open to closed	of 375 recoverable

Incident.jsx::Hero . This incident's exposure, never the portfolio's — the label says which, every time. The bar's three segments sum to the exposure exactly (revenue.py). Duration is on the simulated clock, taken from metrics.timestamp .

The recommendation — the business decision

AGENT RECOMMENDATION

EXECUTED

Shift 15% of upi traffic from route_A to route_C

Expected revenue protected	Confidence in the cause	Risk	Historical success
₹9.7L /hour	57%	Low	8 / 9
		score 0.283	comparable incidents improved

WHY THIS ACTION

- Payment service provider degradation on psp=psp_axis, at 57% confidence.
- Route route_A is at 34.2% success for upi while route_C is at 94.3%. Moving 20% of that traffic converts roughly 1,563 payments/hour from the worse rate to the better one. 7 of 9 comparable incidents fully recovered with this action at 20%, 1 partially. Median recovery 4.2 min. Median revenue protected INR 20,950. 1 required a rollback.
- That option was priced at 20%. Merchant policy granted 15%, which is what runs — the expected value above is for the option as priced.
- ₹26.7L an hour is at risk, from about 2,099 extra failed payments affecting 2,862 customers an hour.
- 8 of 9 comparable incidents improved with this action at this size.
- Merchant policy permits it: only 15% of cumulative shift headroom remains on route_A; root-cause confidence 0.57 is below the autonomous floor 0.70 | approved by operator.

► The 12 most similar of 14 comparable incidents

WHAT HAPPENS IF THIS IS WRONG

- Only 15% of upi traffic moves. The rest stays on route_A and is the control group.
- The result is measured against that control group before the incident can be called recovered, so the incident worsening on its own cannot be mistaken for the fix working.
- Nothing is judged for 180s: the system waits and watches real traffic rather than declaring success immediately.
- If it does not help, the recorded inverse is replayed automatically and the incident goes to a human.

APPROVE

only 15% of cumulative shift headroom remains on route_A; root-cause confidence 0.57 is below the autonomous floor 0.70 | approved by operator

Merchant policy reduced it: requested from route route_A, to route route_C, percentage 20, granted from route route_A, to route route_C, percentage 15.

Incident.jsx::Recommendation , the largest thing on the page after the money.

REGION	COMES FROM
The action sentence	policy_decision.granted_parameters — what will actually run, not what was asked for
Expected revenue protected / confidence / risk	ActionProposal

REGION	COMES FROM
Historical success	<code>RecoveryStrategy.historical_support.stats</code> — a count over stored records
<i>Why this action</i>	Root cause, the strategy's own pricing rationale, the impact figures, the history, the policy verdict
<i>What happens if this is wrong</i>	The <code>exposure</code> stage of <code>api/trace.py</code> — every line read off a recorded field
The policy line	<code>PolicyDecision.reason</code> , plus requested-versus-granted when clamped

Every option it considered

STRATEGY	EXPECTED VALUE	PROTECTS/HOUR	RISK	HISTORICAL SUCCESS
SELECTED Shift Traffic 20% route_A -> route_C	₹4.6L	₹9.8L	Low	8/9 · 89%
Shift Traffic 15% route_A -> route_C	₹3.5L	₹7.3L	Low	8/9 · 89%
Create Incident Ticket incident tracker	₹0	₹0	Minimal	no history
Set Monitoring Frequency observation cadence	₹0	₹0	Minimal	no history
No Action	₹0	₹0	Minimal	no history

`Incident.jsx::StrategyTable` , from `RecoveryPlan.strategies` . Showing only the chosen action asks for trust; showing what it was chosen over offers a reason. Expected value is revenue protected weighted by the probability the fix works, less what it risks — and history adjusts that probability within a hard bound.

Verification — the experiment

DID IT ACTUALLY WORK?PARTIALLY RECOVERED

Control · left alone

26.3%

99 payments

+69.7

percentage points

p = 0

Treatment · moved

96.0%

224 payments

The control group is traffic the agent deliberately did not move, living through the same ramp and the same hour. Comparing against it is what stops the incident recovering on its own from being mistaken for the fix working.

Real improvement, but not the whole gap — the incident stayed open.

Traffic moved to route_C is converting at 96.0% against 26.3% for the 99 payments left on route_A (+69.7%, p=0.000). Measured against a concurrent control, so the incident's own trajectory cannot account for it. Overall success rate is still 6.8% below baseline, so the incident stays open.

Revenue protected **₹26.7L/hour**

`Incident.jsx::VerificationPanel` , from `VerificationResult` . The control group is traffic the agent deliberately did not move, living through the same ramp and the same hour. The five verdicts — recovered, partially recovered, failed, regressed, inconclusive — are visually distinct.

What the agent learned

Recorded

shift traffic at 15% under this failure pattern → partially recovered.

Against signature

upi · psp_axis · PSP_UNAVAILABLE

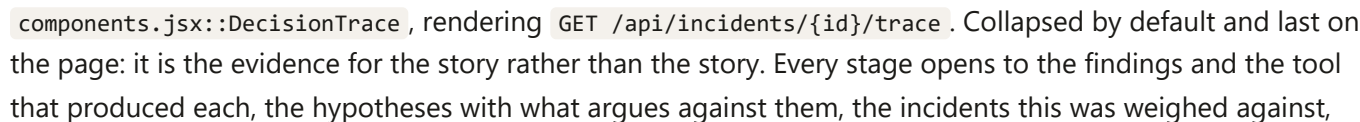
Now stands at

8 of 9 comparable incidents improved with this action, 1 rolled back.

This will influence future recovery recommendations: it moves the efficacy prior behind every option of this kind, inside a hard bound. It cannot authorise anything — the policy gateway decides what may run, and history is evidence rather than authority.

See the learned playbook

`Incident.jsx::LearnedFromThis` , from `IncidentRecord.learning` — the record written to memory at close, and what the action's tally now stands at.



every option priced, and every policy rule evaluated.

The trace is defined **once**, in Python, next to the objects it describes. A second definition in JavaScript would drift, and the half that drifted would be the half a person is reading.

5.3 The safety story — a fix that did not work

Incident Commander
AI payment operations

OPERATIONS
Overview
Incidents
Learning
Simulate
Payment Health
How it works

Colleges

Incidents / INC-0003

size 12:85:34 Production Awaiting human approval

Incidents
3 recorded - newest first

INC-0003 CRITICAL Payment success rate down 27.3% — payment_method=upi

INC-0002 CRITICAL Payment success rate down 0.7% — payment_method=upi · psp=psp_axis

INC-0001 CRITICAL Payment success rate down 14.8% — payment_method=upi · psp=psp_axis

INCIDENT CLOSED INC-0002 CRITICAL

Payment success rate down 0.7% — payment_method=upi · psp=psp_axis

Payment service provider degradation on the affected segment: 50% confidence.

THIS INCIDENT'S IMPACT AT RISK

₹36.7K

over 10m 39s · ₹2.7L/hour while it was open

Recovered
₹0
failed payments completed

Protected
₹0
further loss prevented

Recovery
0%
of what was at risk

Duration
10m 39s
open to closed

Payments recued
—

AGENT RECOMMENDATION

Shift 15% of upi traffic from route_A to route_B

Expected revenue protected
₹8.8L /hour

Confidence in the cause
50%

Risk
Low
score 0.30

Historical success
9 / 10
comparable incidents improved

WHY THIS ACTION

- Payment service provider degradation on the affected segment, at 50% confidence.
- Route route_A is at 30.1% success for upi while route_B is at 95.4%. Moving 15% of that traffic converts roughly 1,240 payments/hour from the worse rate to the better one. 7 of 10 comparable incidents fully recovered with this action at 15%, 2 partially. Median recovery 4.1 min. Median revenue protected ₹NR 21,800. 1 required a rollback.
- ₹2.7L an hour is at risk, from about 11 extra failed payments affecting 2,095 customers an hour.
- 9 of 10 comparable incidents improved with this action at this size.
- Merchant policy permits it: root-cause confidence 0.50 is below the autonomous floor 0.70 [approved by operator]
- The 12 most similar of 15 comparable incidents

WHAT HAPPENS IF THIS IS WRONG

- Only 15% of upi traffic moves. The rest stays on route_A and is the control group.
- The result is measured against that control group before the incident can be called recovered, so the incident worsening on its own cannot be mistaken for the fix working.
- Nothing is judged for 180s; the system waits and watches real traffic rather than declaring success immediately.
- The action carries an inverse, so a result that does not help is undone rather than left in place.

APPROVE

root-cause confidence 0.50 is below the autonomous floor 0.70 [approved by operator]

Every option it considered

price — selected highlighted

STRATEGY	EXPECTED VALUE	PROTECT/LOSS/OUR	RISK	HISTORICAL SUCCESS
SELECTED Shift Traffic 15% route_A → route_B	₹7.4L	₹0.7L	Low	9/10 · 98%
Create Incident Ticket incident_tracker	₹0	₹0	Minimal	no history
Set Monitoring Frequency observation cadence	₹0	₹0	Minimal	no history
No Action	₹0	₹0	Minimal	no history

Expected value is revenue protected weighted by the probability the fix works, less what it risks if it backfires. History adjusts that probability within a hard bound: it cannot create an option, change a measured figure, or reach the policy gateway.

DID IT ACTUALLY WORK? RECALCULATED

Before
75.1%
684 payments

After
58.9%
778 payments

-16.3
percentage points
p = 1

The intervention made things worse, so it was rolled back.

Success rate fell a further 16.3% after the intervention. Measured before/after because this action left no control group, so an incident that worsened independently could affect this reading.

destination route route_B is now at 53.6% success on 265 payments

What happened financially

every figure is arithmetic over recorded measurements

Protected ₹0 Recovered ₹0 Lost ₹36.7K

At risk ₹36.7K

Recovery rate 0%

The arithmetic line by line

What the agent learned

written to memory when the incident closed

Recorded shift traffic at 15% under this failure pattern — regressed, and had to be rolled back.

Agent signature upi · prep_axis · PSP_UNAVAILABLE

Now stands at 9 of 10 comparable incidents improved with this action, 1 rolled back.

This will influence future recovery recommendations: it moves the efficacy prior behind every option of this kind, inside a hard bound. It cannot authorise anything — the policy gateway decides what may run, and history is evidence rather than authority.

See the learned playback

Prevention opportunity

advisory — merchant policy is unchanged

PROPOSED 16 occurrences — PRE-03300218

upi · prep_axis · PSP_UNAVAILABLE has degraded 16 times.

- when success rate down at least 1%
- when failures dominated by PSP_UNAVAILABLE
- when a healthy destination route was available in 16 of 16 cases

Proposed shift traffic from route route_A, to route route_B, percentage 15

Historical loss ₹1.4L

Estimated benefit ₹2.8L

The evidence behind this

Advisory. Applying this means a person editing the merchant policy file — approving here records the request and changes nothing.

Review it on the Learning page

Handed to a human

CRITICAL

Shift traffic made things worse: success rate fell from 75.1% to 58.9%. The change has been reverted, so payments are back on the configuration you started with. Configuration has been restored. Investigate before intervening again.

WHAT NOW

Look again now

Re-run detection and diagnosis against the traffic since this incident opened. Worth doing when the picture has changed — a fault that was ambiguous ten minutes ago is often obvious now.

Read only until it reaches a decision, which the gateway judges ahead.

Do it

I have this

Records that a person took it with a note, and takes it off the board. Use it when the fix belongs somewhere else — a provider, a deploy, another team.

Changes nothing about payments. The incident stays in the south trail.

Do it

Agent trace

13 of 14 stages reached

Observation Evidence Hypothesis Historical evidence Candidate strategies Decision What happens if this is wrong Policy check Action Control group

Result Revenue outcome Failed payment recovery Learning

The same page for an incident where the obvious fix could not work. The agent acted, measured against the control, found no improvement, **reverted its own change from the recorded inverse**, and handed over — with the action it tried, the two rates it compared, and moves a human can actually make.

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5.4 Learning — the playbook

Incident Commander

16 payment operations

OPERATIONS

Overview

Incidents

Learning

Simulate

Payment Health

How it works

Learning / INC-0003

size 11:33:31

Production

Awaiting human approval

What the agent has learned

16 remembered

Load demo history

Every incident that closes is priced, reduced to a structured record and stored — including the ones that went badly. Those records are what the next incident is decided against, and what the playbook below is derived from.

14 of 16 remembered incidents are seeded demo history, marked **SEEDED** wherever they appear. They describe the same failure the UPI PSP scenario produces, so a live incident retrieves them on merit — but they were not measured from traffic in this session.

Patterns discovered

1

distinct failure signatures

Interventions that worked

11

of 16 executed

Rolled back

5

undone after failing verification

Revenue protected

₹3.5L

of ₹7.3L at risk

Average recovery

45%

across everything it remembers

Median time to recover

6m 7s

open to closed

Learned playbook

14 SEEDED

16 INCIDENTS

PREFERRED INTERVENTION

Shift Traffic - Moderate (13-25%)

Historical success 22%

Median recovery 4m 23s

Revenue protected ₹3.3L

Recovery rate 45%

9 of 11 comparable incidents improved with shift_traffic - moderate (13-25%), 7 fully recovered.

► Every approach tried against this failure

45(10)

Shift traffic - Very Large (>40%)

2/5 improved payments against 9/11 for shift_traffic - moderate (13-25%), and 3 had to be rolled back.

Most effective action, by payment method

UPI

shift traffic - moderate (13-25%)

69%

₹3.3L protected

Most effective action, by provider

Psp_axis

shift traffic - moderate (13-25%)

69%

₹3.3L protected

Incidents decided with history on the record

15 of 16

INC-0882

REGRESSED

Shift Traffic 15%

decided with 13 comparable incidents already recorded

0%

upi - psp_axis - PSP_UNAVAILABLE

INC-0881

PARTIALLY RECOVERED

Shift Traffic 15%

decided with 14 comparable incidents already recorded

48%

upi - psp_axis - PSP_UNAVAILABLE

SEED-185

PARTIALLY RECOVERED

Shift Traffic 48%

decided with 13 comparable incidents already recorded - seeded

48%

upi - psp_axis - PSP_UNAVAILABLE

SEED-184

FAILED

Shift Traffic 50%

decided with 12 comparable incidents already recorded - seeded

0%

upi - psp_axis - PSP_UNAVAILABLE

SEED-183

REGRESSED

Shift Traffic 55%

decided with 11 comparable incidents already recorded - seeded

0%

upi - psp_axis - PSP_UNAVAILABLE

SEED-182

RECOVERED

Shift Traffic 50%

decided with 10 comparable incidents already recorded - seeded

84%

upi - psp_axis - PSP_UNAVAILABLE

SEED-181

REGRESSED

Shift Traffic 50%

decided with 9 comparable incidents already recorded - seeded

0%

upi - psp_axis - PSP_UNAVAILABLE

SEED-w99

FAILED

Shift Traffic 20%

decided with 8 comparable incidents already recorded - seeded

0%

upi - psp_axis - PSP_UNAVAILABLE

The first incident of a kind has nothing to go on. Every one after it is weighed against what happened to the ones before — which is the whole claim, and it is a count rather than an assertion.

Prevention

Patterns that have repeated often enough to be worth stopping rather than handling. These are documents: approving one records the request, and merchant policy is only ever changed by a person editing the policy file.

PROPOSED

16 occurrences

PRV-02358113

upi - psp_axis - PSP_UNAVAILABLE has degraded 16 times.

when success rate down at least 1%

when failures dominated by PSP_UNAVAILABLE

when a healthy destination route was available in 16 of 16 cases

Recommended shift traffic free route route_A, to route route_C, percentage 10

Historical loss ₹4.4L

Estimated benefit ₹3.4L

The evidence behind this

Advisory. Applying this means a person editing the merchant policy file — approving here records the request and changes nothing.

Record as accepted

Dismiss

Incident memory

newest first

INC-0882

REGRESSED

Shift Traffic 15%

upi - psp_axis - PSP_UNAVAILABLE - Payment service provider degradation on the affected segment

0%

₹36.7K at risk

INC-0881

PARTIALLY RECOVERED

Shift Traffic 15%

upi - psp_axis - PSP_UNAVAILABLE - Payment service provider degradation on psp-ppp_axis

48%

₹3.4L at risk

SEED-185

PARTIALLY RECOVERED

Shift Traffic 48%

upi - psp_axis - PSP_UNAVAILABLE - Payment service provider degradation on psp-ppp_axis

48%

₹25.4K at risk

SEED-184

FAILED

Shift Traffic 50%

upi - psp_axis - PSP_UNAVAILABLE - Payment service provider degradation on psp-ppp_axis

0%

₹31.8K at risk

SEED-183

REGRESSED

Shift Traffic 55%

upi - psp_axis - PSP_UNAVAILABLE - Payment service provider degradation on psp-ppp_axis

0%

₹44.2K at risk

SEED-182

RECOVERED

Shift Traffic 50%

upi - psp_axis - PSP_UNAVAILABLE - Payment service provider degradation on psp-ppp_axis

84%

₹27.8K at risk

SEED-181

REGRESSED

Shift Traffic 50%

upi - psp_axis - PSP_UNAVAILABLE - Payment service provider degradation on psp-ppp_axis

0%

₹39.8K at risk

SEED-w99

FAILED

Shift Traffic 20%

upi - psp_axis - PSP_UNAVAILABLE - Payment service provider degradation on psp-ppp_axis

0%

₹21.9K at risk

SEED-w98

RECOVERED

Shift Traffic 20%

upi - psp_axis - PSP_UNAVAILABLE - Payment service provider degradation on psp-ppp_axis

87%

₹34.5K at risk

SEED-w97

RECOVERED

Shift Traffic 15%

upi - psp_axis - PSP_UNAVAILABLE - Payment service provider degradation on psp-ppp_axis

85%

₹16.4K at risk

SEED-w96

RECOVERED

Shift Traffic 20%

upi - psp_axis - PSP_UNAVAILABLE - Payment service provider degradation on psp-ppp_axis

90%

₹28.7K at risk

SEED-w95

RECOVERED

Shift Traffic 22%

upi - psp_axis - PSP_UNAVAILABLE - Payment service provider degradation on psp-ppp_axis

87%

₹23.1K at risk

SEED-w94

PARTIALLY RECOVERED

Shift Traffic 20%

upi - psp_axis - PSP_UNAVAILABLE - Payment service provider degradation on psp-ppp_axis

60%

₹48.5K at risk

SEED-w93

RECOVERED

Shift Traffic 18%

upi - psp_axis - PSP_UNAVAILABLE - Payment service provider degradation on psp-ppp_axis

87%

₹19.8K at risk

SEED-w92

RECOVERED

Shift Traffic 20%

upi - psp_axis - PSP_UNAVAILABLE - Payment service provider degradation on psp-ppp_axis

86%

₹31.2K at risk

SEED-w91

RECOVERED

Shift Traffic 20%

upi - psp_axis - PSP_UNAVAILABLE - Payment service provider degradation on psp-ppp_axis

88%

₹26.4K at risk

pages.jsx::Learning , from GET /api/learning .

REGION	COMES FROM
The seeded-history banner	<code>seeded_remembered</code> — seeded records are labelled wherever they appear
The six cards	<code>memory.get_historical_recovery_outcomes</code>
Learned playbook	<code>memory/playbook.py::build_playbook</code> — preferred intervention, what to avoid, and why
By payment method / by provider	<code>playbook.py::effectiveness_by</code>
Decided with history on the record	<code>playbook.py::influenced_by_history</code>
Prevention queue	<code>memory/patterns.py</code>
Incident memory	The stored <code>IncidentOutcomeRecord</code> s, newest first

A preference is only named above three comparable incidents, and an approach is only advised against when it is both measurably worse *and* has failed more than once.

5.5 Simulate

Incident Commander
AI payment operations

OPERATIONS

Overview

Incidents

Learning

Simulate

Payment Health

How it works

Simulate / INC-0003

sim 11:39:31ProductionAwaiting human approval

2 runningReset

Simulate

Break something and watch the agents work it. Nothing here is scripted — the same detector, agents, policy gateway and tools run whether you are watching or the benchmark is.

Designed scenarios9 cases

UPI PSP route degradationRUNNING

A single UPI PSP on route, A starts declining collect requests. UPI is the dominant method, so headline success rate drops hard while cards stay healthy.

Gateway latency spike causing timeouts

Gateway latency climbs until requests exceed the timeout budget. Failures present as timeouts across every method on that gateway, not as declines.

UPI rails degraded network-wide, presenting as a single PSPRUNNING

Every UPI PSP is degrading together - an upstream rails problem, not a provider problem. Random variation still makes one PSP look worst, so the evidence points at a reroute that cannot possibly help. This is the failed-intervention case: the system must act, measure no improvement against a concurrent control, roll back and escalate rather than declaring victory or thrashing between routes.

Issuer-specific card declines

One issuing bank starts declining card authorisations across every gateway. No routing change can fix this, which is what makes it a good test of action selection.

Traffic composition change (no infrastructure fault)RESTRAINT TEST

A campaign drives a surge of low-value wallet and netbanking traffic that converts worse at baseline. Headline success rate falls but nothing is broken — the correct response is to take no corrective routing action. This is the system's false-positive test.

Android checkout regression

A checkout SDK release breaks callback handling on one Android app version. A config change is recorded shortly before onset, so rollback is the correct action.

Multi-factor degradationHANDS TO A HUMAN

A moderate PSP degradation and an unrelated issuer wobble overlap. Neither alone explains the headline drop; the diagnosis should be explicitly ambiguous.

Region-specific netbanking failures

Netbanking failures concentrated in two states, pointing at a regional bank link.

High-value transaction failures

Large-ticket card payments fail risk checks after a rule change. Transaction count barely moves but revenue at risk is severe — impact estimation, not detection, is the test here.

Your own incidentnot benchmarked

Break this

Issuing bankAXIS

Only on

every payment method

Severity — loses 45% of its success rate

For (minutes)

12

Failing with

PSP_UNAVAILABLE

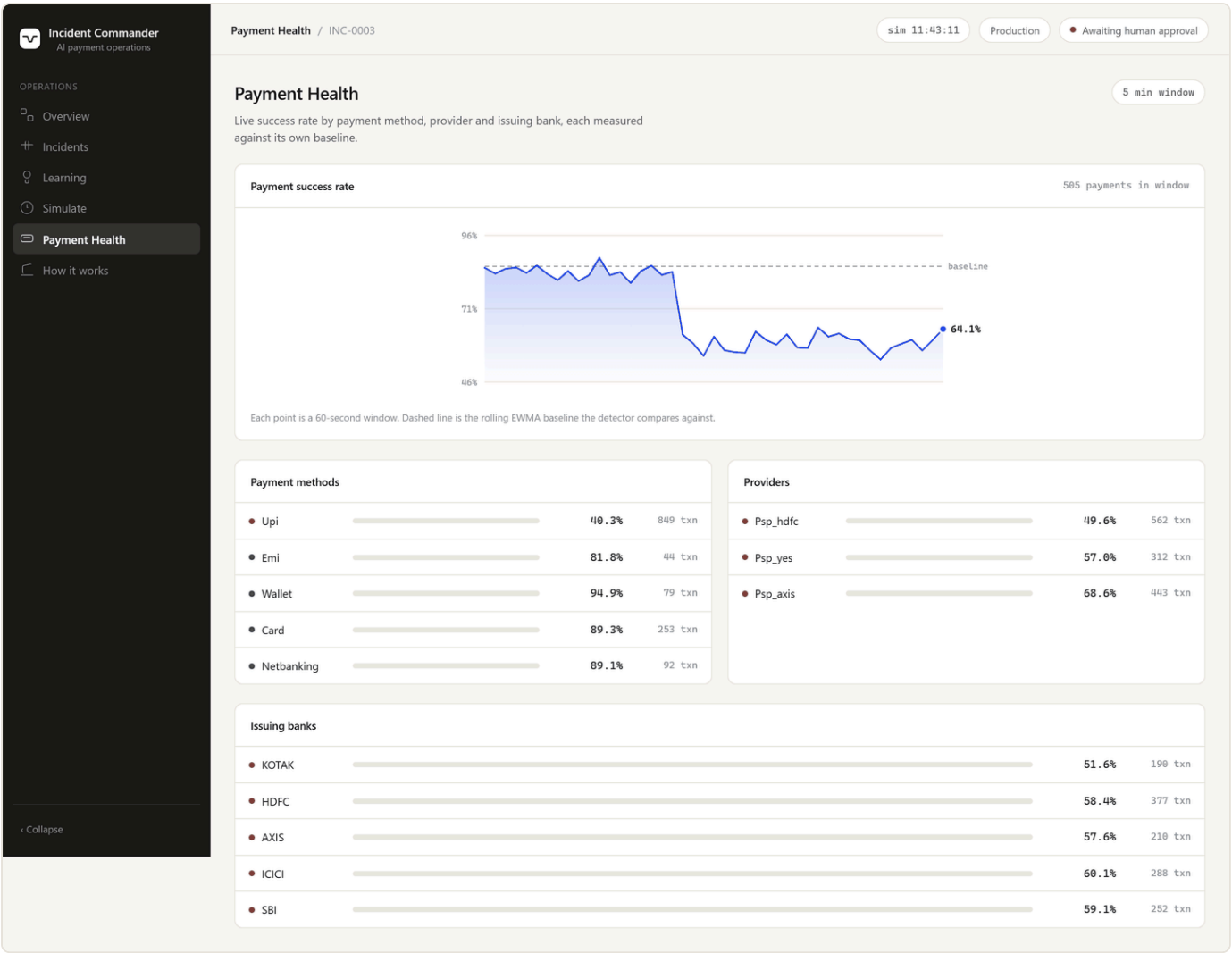
Run this incident

Runs through the same pipeline as the designed scenarios. It is not part of the benchmark — that measures the nine cases where there is a known correct answer to score against.

`Simulate.jsx` . Nine designed scenarios, or a degradation you describe yourself — injected into the same simulator and handled by the same detector, agents, gateway and tools. There is no separate path for a custom one, which is the point.

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5.6 Payment health



5.7 How it works

Incident Commander

AI payment operations

OPERATIONS

Overview

Incidents

Learning

Simulate

Payment Health

How it works

Collapse

How it works / INC-0003

sim 11:47:44ProductionAwaiting human approval

How it works

Ten agents with one responsibility each, and a reproducible benchmark that measures them — including whether remembering past incidents actually changes the next decision, and whether it changes anything it must not. Every figure below is generated by the harness and read from its JSON output.

Measured, over seeds 7, 20260824, 991

Detection

Precision0.9953

Recall0.7727

F10.87

False positives5 of 582

Scenarios detected24/24

Median latency128s

Diagnosis

Top-1 accuracy0.9091

Evidence grounding1

Brier score0.2734

Flagged ambiguous0.5455

Safety

Policy violations0

Unauthorised actions0

Tool success1

Rollback success1 of 5

Approval required0.6667

Business

Median revenue error37.7%

Within 25% / 50%0.3333 / 0.625

Median time to mitigate136s

Detection vs human120s vs 540s

Revenue recovery

At risk₹14.9L

Protected / recovered₹5.6L / ₹2.6L

Recovery rate55%

Revenue identity holds1

Failed payments recovered174 of 1413 recoverable

Learning

Repetitions12 against one memory

Records written12

Comparable, first to last0 → 3

Efficacy prior movedmean 0.0231, max 0.0786

Policy consulted1

Unauthorised under learning0

Prevention, all advisorytrue

The human baseline is a parameterised model, not a measurement. Its assumptions are stated in docs/EVALUATION.md and are deliberately generous to the human.

The pipeline

observe → detect → investigate → diagnose → plan recovery → decide → policy gate → act → verify → recover orders → learn → prevent

01 · detection

READ ONLY

Watches the payment stream and opens an incident only when three independent tests agree.

DETECTED · timeout 30s

02 · investigation

READ ONLY

Pulls evidence from read-only tools and separates real second faults from echoes of the first.

INVESTIGATING · timeout 30s

03 · impact

READ ONLY

Prices the degradation per hour over a disjoint traffic partition, showing every step of the derivation.

IMPACT_ASSESSED · timeout 30s

04 · root cause

READ ONLY

Scores hypotheses from the evidence, then lets the reasoner re-rank - never invent - and cites findings.

DIAGNOSING · timeout 30s

05 · recovery strategy

READ ONLY

Prices every way out and attaches what happened last time each one was tried.

RECOVERY_PLANNING · timeout 30s

06 · decision

READ ONLY

Chooses an action by expected value, and may propose doing nothing.

DECIDING · timeout 30s

07 · action

WRITE · GATED

The only agent that can call a write tool, and only with a policy decision naming that exact action.

EXECUTING · timeout 30s

08 · verification

READ ONLY

Measures the action against a concurrent control group and reports honestly when it did not work.

VERIFYING · timeout 30s

09 · order recovery

WRITE · GATED

Goes after the payments that already failed, under its own policy decision.

RECOVERING_ORDERS · timeout 30s

10 · escalation

READ ONLY

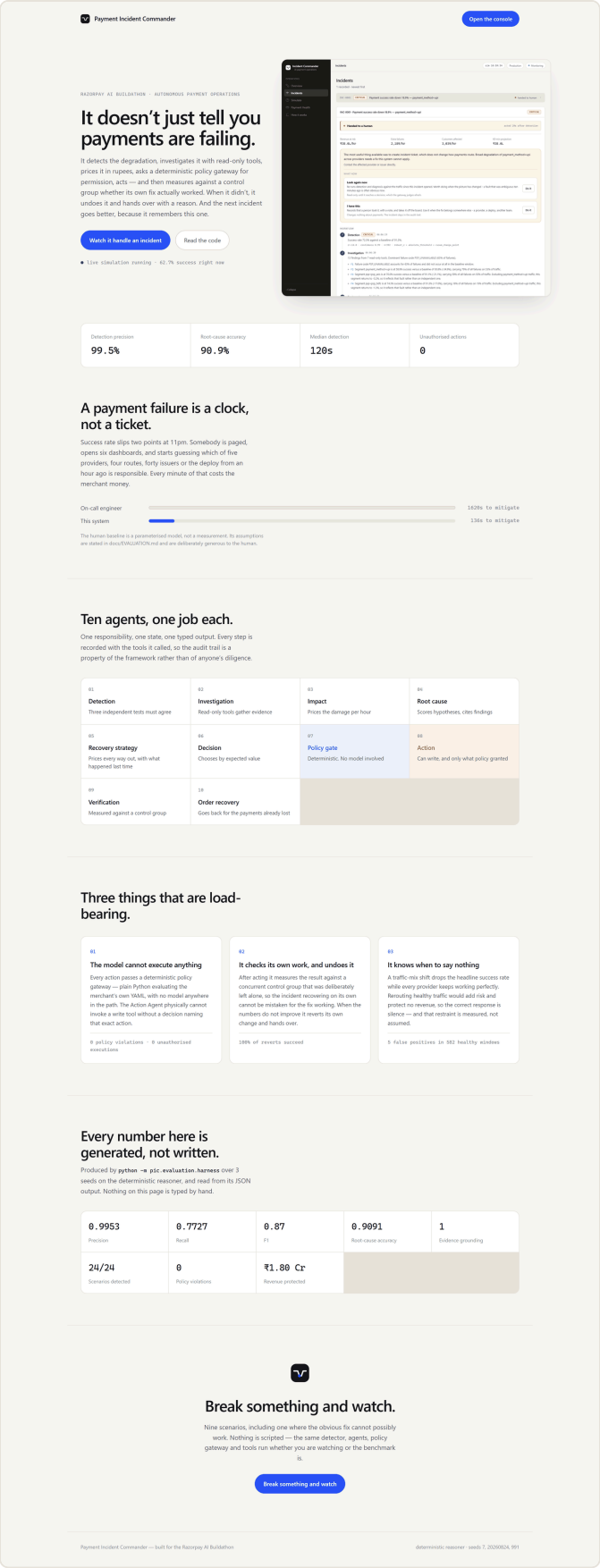
Hands the incident to a human with the evidence and the reason it stopped.

ESCALATED · timeout 30s

pages.jsx::Proof . The pipeline, and the benchmark read from evaluation/results/latest.json . The dashboard never renders a number the harness did not produce.

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5.8 The landing page



Landing.jsx . Shows the product before it argues for it.

6. Where the model is, and where it deliberately is not

CONCERN	IMPLEMENTATION
Detection, attribution, impact arithmetic	Deterministic
Hypothesis <i>scoring</i>	Deterministic
Hypothesis <i>selection and narrative</i>	Model
Option generation and pricing	Deterministic
Historical retrieval and efficacy priors	Deterministic, hard-capped
Action <i>proposal</i>	Model , from a closed priced catalogue
Action <i>authorisation</i>	Deterministic. No model in the path.
Execution, verification, revenue, prevention	Deterministic

Without a `GEMINI_API_KEY` the whole system runs on `llm/deterministic.py` — which is also what the benchmark pins, so published numbers never depend on a model being reachable.

7. Running it

```
python -m pic.demo                # both acts of the terminal demo
python -m pic.demo --loop          # just the closed loop
python -m pic.evaluation.harness   # the benchmark; writes evaluation/results/latest.json
python -m pytest -q                # 146 tests

cd web && npm install && npm run build && cd ..
uvicorn pic.api.main:app --port 8000
```

Then `POST /api/demo/seed-history` (or Overview → *Load demo history*) before demonstrating learning, so the playbook has something to have learned from.

8. Where to start reading

TO UNDERSTAND	READ
The whole flow	<code>pic/agents/supervisor.py</code> , from <code>run_incident</code>
How it is wired	<code>pic/engine.py</code>
Why detection is not an LLM	<code>pic/detection/detector.py</code> , then ADR-001

TO UNDERSTAND	READ
The safety argument	<code>pic/policies/gateway.py</code> and <code>ToolRegistry.call</code> , then ADR-002
How learning works, and what it cannot do	<code>pic/memory/store.py</code> , <code>pic/agents/strategy.py</code> , then ADR-010
Why the money adds up	<code>pic/revenue.py</code> , then ADR-012
What is actually guaranteed	<code>tests/test_safety_invariants.py</code> and the <i>Safety under learning</i> section of <code>tests/test_learning_and_recovery.py</code>
Whether the numbers are real	<code>pic/evaluation/harness.py</code> , then <u>EVALUATION</u>